Data Inventory & Classification

Ref: 4.1
Product: iConnect Plus
Organisation: Cache Digitech
Classification: Internal — Confidential
Version: 2.0
Date: 2026-11-03
Deployment model: On-premises (Cache Digitech premises) · VPN for remote access
Audience: DPO/designated officer, security assessors, module leads, auditors

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1. Purpose

Inventory **categories of data** processed by iConnect Plus on **Cache Digitech premises**, assign **classification labels**, and map storage locations on the on-prem stack (PostgreSQL, MinIO, Redis, logs). This supports DPDP accountability, access control design, and encryption scope.

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2. Classification scheme

Aligned with `apps/api/src/modules/platform/data_classification.py`:

Level	Code	Description	Handling
Public	`public`	Non-sensitive; may appear in marketing or open docs	Standard access controls
Internal	`internal`	Business operational; not for external disclosure	RBAC; LAN/VPN only
Confidential	`confidential`	PII, customer/vendor, HR, audit metadata	RBAC + encryption where inventoried; need-to-know
Restricted	`restricted`	Credentials, tokens, payroll, bank, high-impact finance	Strong RBAC, Fernet fields, no plaintext logging

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3. Data stores (on-premises)

Store	Technology	Volume / path (typical)	Primary contents	Location
OLTP	PostgreSQL 16 (`erp-postgres`)	`erp_pgdata`	All tenant business entities	India — premises
Objects	MinIO (`erp-minio`)	`erp_minio_data`	Attachments, exports, document binaries	India — premises
Cache / sessions	Redis (`erp-redis`)	`erp_redis_data`	Permission cache, Celery results (non-authoritative)	India — premises
Messages	RabbitMQ	`erp_rabbitmq_data`	Async job envelopes	India — premises
Upload scratch	Docker volumes	`erp_crm_uploads`, `erp_asset_storage`	Transient upload paths	India — premises
Application logs	Host / container logs	**Template / Pending sign-off** path	Security & ops events	India — premises (≥180d target)

No **primary** business OLTP or object store is hosted on public cloud in this deployment model.

4. Representative data inventory

Data domain	Examples	Typical class	System of record	Encrypted fields (where applicable)
Identity & access	Users, roles, sessions, refresh tokens	Confidential / Restricted	`foundation.*`	`password_hash`, token hashes, selected secrets
Organization	Tenants, companies, branches	Internal / Confidential	`organization.*`	—
HR & payroll	Employee, salary, tax IDs	Confidential / Restricted	`hr.*`, `payroll.*`	PAN, Aadhaar, bank details per inventory
Finance	GL, journals, bank accounts	Confidential / Restricted	`finance.*`	Bank account numbers, tax identifiers
Master data	Customers, vendors, products	Internal / Confidential	`master_data.*`	Tax/bank where captured
Documents	Files, versions, retention	Confidential	PostgreSQL + MinIO	—
Audit	Immutable audit log	Confidential	`audit.audit_log`	—
Notifications	Email/push event records	Internal	`notification.*`	—
Platform	Outbox, idempotency, job runs	Internal	`platform.*`	—

Full table-level map: extend `TABLE_CLASSIFICATION` in code; export for audit: **Template / Pending sign-off**.

5. Code-registered high-sensitivity tables

Table key	Class
`foundation.sec_user`	confidential
`foundation.sec_session`	restricted
`foundation.sec_refresh_token`	restricted
`payroll.pay_employee_salary`	restricted
`finance.fin_bank_account`	restricted
`master_data.master_vendor`	confidential
`master_data.master_customer`	confidential
`audit.audit_log`	confidential

6. Encrypted field inventory (must not log in plaintext)

From `ENCRYPTED_FIELD_INVENTORY`: `password_hash`, `refresh_token_hash`, `bank_account_number`, `tax_id`, `tax_number`, `pan`, `pan_number`, `aadhaar`, `api_secret`, `oauth_client_secret`, `payment_token`, `ssn`, `bank_details_json`, `secret_vault_ref`, `client_secret_vault_ref`, and extensions per module.

7. Data flows (summary)

Flow	Path	Classification impact
User browser → app	LAN or VPN → TLS (nginx) → web/API	All classes in transit protected by TLS
API → PostgreSQL	Docker network `erp-net`	At-rest: DB volume encryption **Template / Pending sign-off** (OS/disk)
API → MinIO	S3-compatible internal endpoint	Bucket policy + credentials in `.env`
Optional email/SMS	SMTP/Brevo, FCM	Message content may include confidential data — processor agreements **Template / Pending sign-off**
Backups	pg_dump / volume snapshot → on-prem target	Same classification as source — see 4.4

8. Roles & responsibilities

Role	Responsibility
Data owner (business)	Defines retention and lawful basis per domain
Platform engineering	Maintains classification registry in code
IT	Physical/logical security of premises hosts
Tenant admin	User provisioning within tenant boundary

9. Related documents

Ref	Document
1.2	Data Flow Diagram
4.2	Cryptography & Key Management (CBOM)
4.3	Data Retention & Deletion Schedule
4.5	Personal Data Processing Record
3.1	RBAC Matrix

10. Change log

Date	Change	By
2026-11-03	v2.0 inventory baseline	Platform

Last reviewed: Template / Pending sign-off